

Question_5:

1. Verilog code :

- Module code:

```
1  module FIFO(din_a,wen_a,ren_b,clk_a,clk_b,rst,dout_b,full,empty);
2  parameter FIFO_WIDTH=16;
3  parameter FIFO_DEPTH=512;
4  input clk_a,clk_b,rst;
5  input wen_a,ren_b;
6  output reg [FIFO_WIDTH-1:0] dout_b;
7  input [FIFO_WIDTH-1:0] din_a;
8  output full,empty;
9  //////////////////////////////////
10 reg [9:0] rd_ptr,wr_ptr;
11 reg [FIFO_WIDTH-1:0] fifo [FIFO_DEPTH-1:0];
12 //////////////////////////////////
13 assign full = ((wr_ptr[8:0] == rd_ptr[8:0])&&(wr_ptr[9]!=rd_ptr[9])) ?1'b1:1'b0;
14 assign empty = (wr_ptr == rd_ptr)?1'b1:1'b0;
15 always @(posedge clk_a ) begin
16     if(rst) begin
17         wr_ptr<=0;
18     end
19     else if ((wen_a)&&(!full)) begin
20         fifo[wr_ptr[8:0]]<=din_a;
21         wr_ptr<=wr_ptr+1'd1;
22     end
23 end
24
25 always @(posedge clk_b) begin
26     if(rst) begin
27         rd_ptr<=0;
28         dout_b<=0;
29     end
30     else if ((ren_b)&&(!empty)) begin
31         dout_b<=fifo[rd_ptr[8:0]];
32         rd_ptr<=rd_ptr+1'd1;
33     end
34 end
35 endmodule
36
```

- Test bench:

```

1  module fifo_tb();
2  parameter FIFO_WIDTH=16;
3  parameter FIFO_DEPTH=512;
4  reg clk_a,clk_b,rst;
5  reg wen_a,ren_b;
6  wire [FIFO_WIDTH-1:0] dout_b;
7  reg [FIFO_WIDTH-1:0] din_a;
8  wire full,empty;
9  //////////////////////////////////
10 FIFO fifo (din_a,wen_a,ren_b,clk_a,clk_b,rst,dout_b,full,empty);
11 //////////////////////////////////
12 initial begin
13     clk_a=0;
14     clk_b=0;
15     forever
16     begin
17         #1 clk_a=~clk_a; clk_b=~clk_b;
18     end
19 end
20 initial begin
21     $readmemh("mem.dat",fifo.fifo);
22     rst=1;
23     wen_a=1;
24     ren_b=0;
25     din_a=16'd20;
26     @(negedge clk_a);
27     rst=0;
28     wen_a=0;
29     ren_b=1;
30     repeat(2) begin
31         @(negedge clk_a);
32     end
33     wen_a=1;
34     ren_b=0;
35     din_a=16'd20;
36     repeat(512) begin
37         //din_a=$random;
38     @(negedge clk_a);
39     end
40     din_a=16'd5;
41     @(negedge clk_a);
42     wen_a=0;

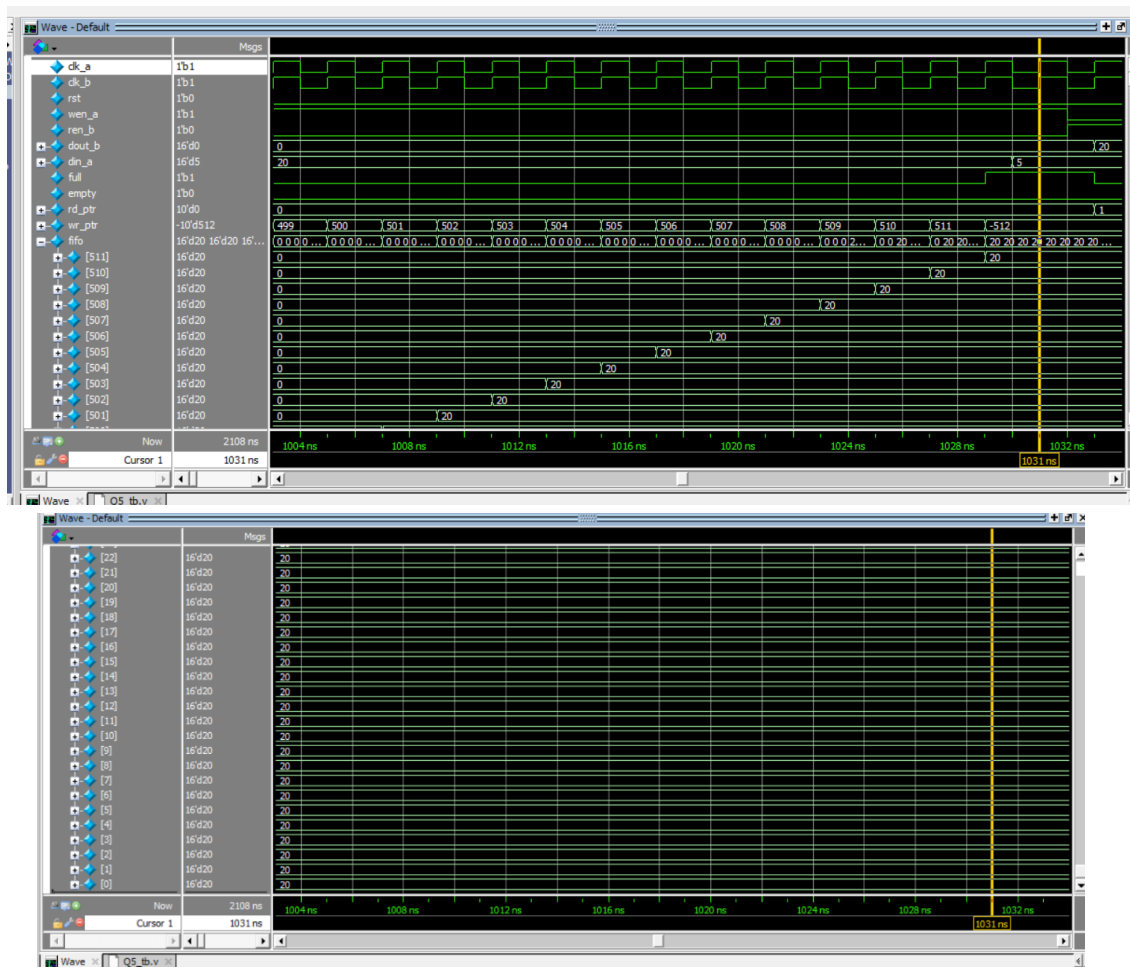
```

```

43     ren_b=1;
44     repeat(12) begin
45         @(negedge clk_a);
46     end
47     wen_a=1;
48     ren_b=0;
49     repeat(13) begin
50         din_a=$random;
51     @(negedge clk_a);
52     end
53     wen_a=0;
54     ren_b=1;
55     repeat(513) begin
56         din_a=$random;
57     @(negedge clk_a);
58     end
59     $stop;
60 end
61 endmodule
62

```

1. Waveform snippet: Waveform:

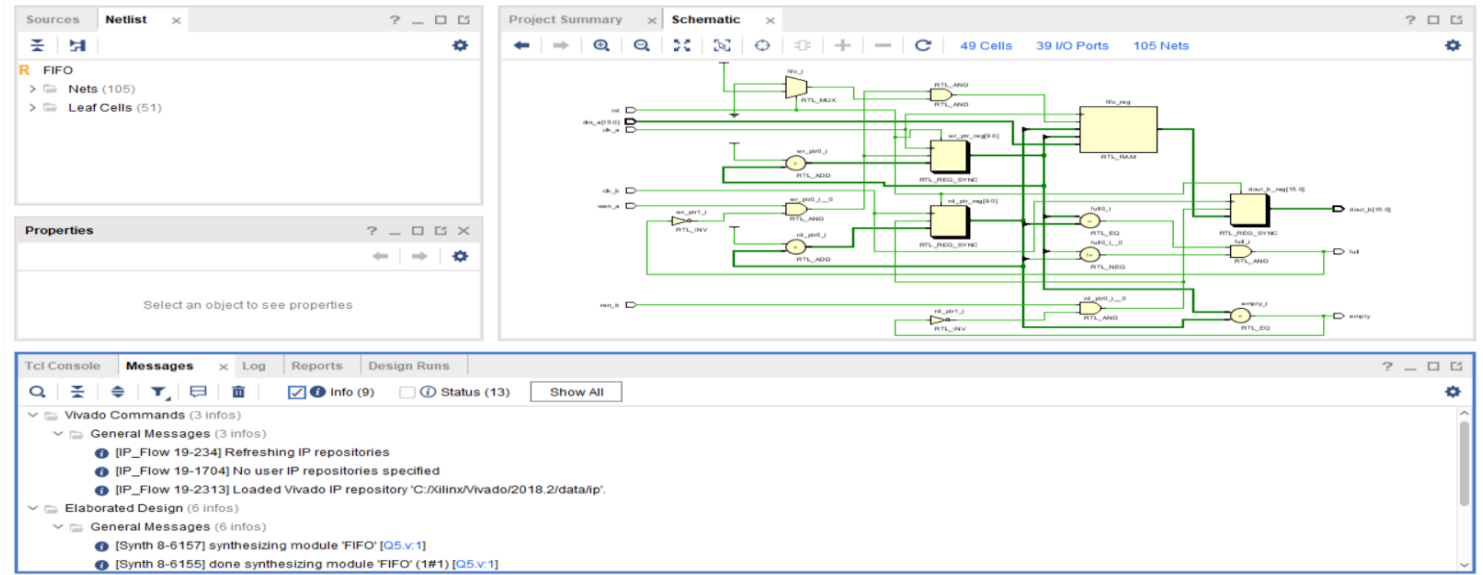


- Do file:

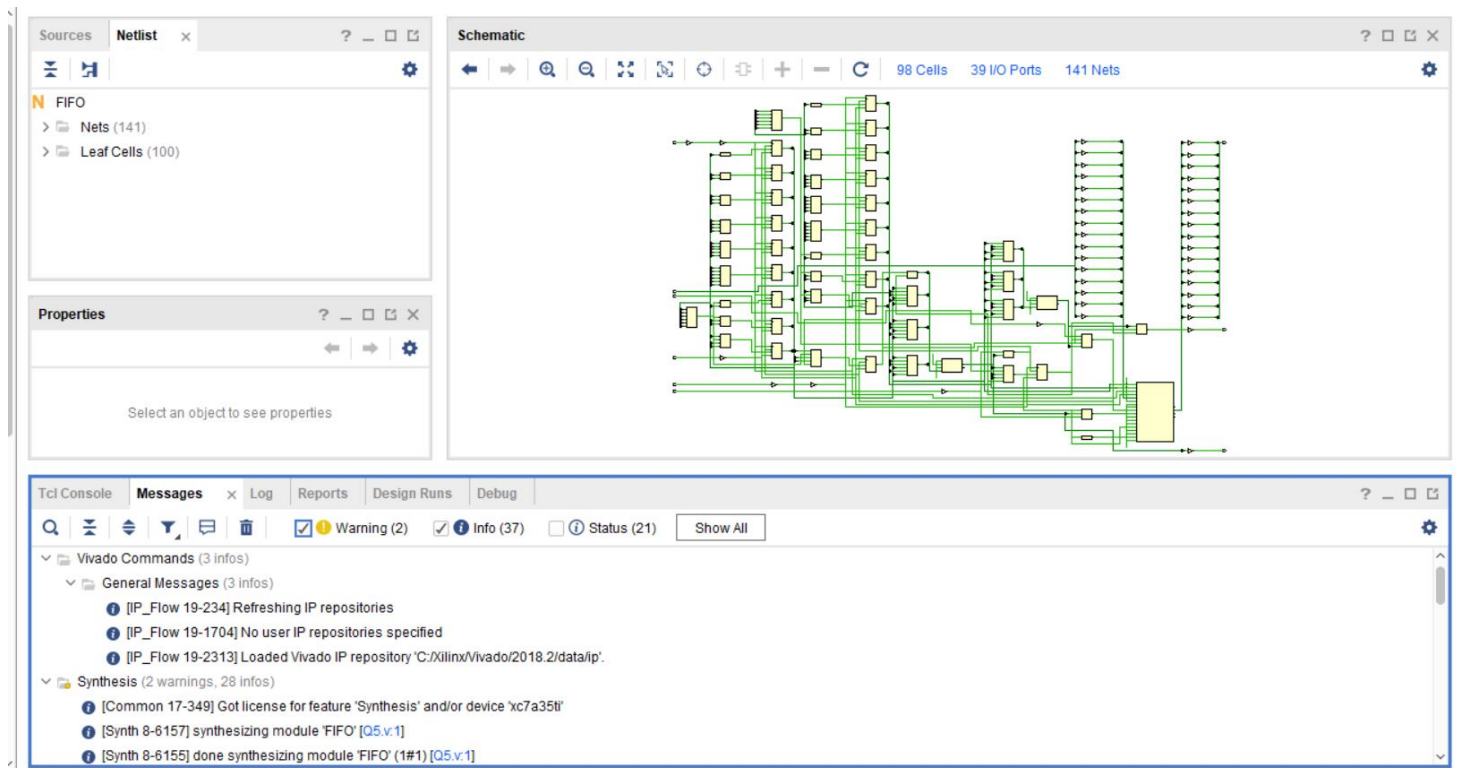
```
vlib work
vlog Q5.v Q5_tb.v
vsim -voptargs=+acc work.fifo_tb
add wave *
run -all
#quit -sim
```

2. Synthesis tool:

- Elaboration:



- Synthesis :



Setup	Hold	Pulse Width
Worst Negative Slack (WNS): 5.483 ns	Worst Hold Slack (WHS): 0.146 ns	Worst Pulse Width Slack (WPWS): 4.500 ns
Total Negative Slack (TNS): 0.000 ns	Total Hold Slack (THS): 0.000 ns	Total Pulse Width Negative Slack (TPWS): 0.000 ns
Number of Failing Endpoints: 0	Number of Failing Endpoints: 0	Number of Failing Endpoints: 0
Total Number of Endpoints: 30	Total Number of Endpoints: 30	Total Number of Endpoints: 12

All user specified timing constraints are met.

Name	Slice LUTs (20800)	Slice Registers (41600)	Block RAM Tile (50)	Bonded IOB (106)	BUFGCTRL (32)
N FIFO	26	20	0.5	39	2

Netlist:

```

13 `timescale 1 ps / 1 ps
14
15 (* FIFO_DEPTH = "512" *) (* FIFO_WIDTH = "16" *)
16 (* STRUCTURAL_NETLIST = "yes" *)
17 module FIFO
18     (din_a,
19      wen_a,
20      ren_b,
21      clk_a,
22      clk_b,
23      rst,
24      dout_b,
25      full,
26      empty);
27     input [15:0] din_a;
28     input wen_a;
29     input ren_b;
30     input clk_a;
31     input clk_b;
32     input rst;
33     output [15:0] dout_b;
34     output full;
35     output empty;
36
37     wire \<const0> ;
38     wire \<const1> ;
39     wire clk_a;
40     wire clk_a_IBUF;
41     wire clk_a_IBUF_BUFG;
42     wire clk_b;
43     wire clk_b_IBUF;
44     wire clk_b_IBUF_BUFG;
45     wire [15:0] din_a;
46     wire [15:0] din_a_IBUF;
47     wire [15:0] dout_b;
48     wire [15:0] dout_b_OBUF;
49     wire empty;
50     wire empty_OBUF;
51     wire empty_OBUF_inst_i_1_n_1;
52     wire empty_OBUF_inst_i_1_n_2;
53     wire empty_OBUF_inst_i_1_n_3;

```

- Constrains file:

```

4  ## Rename the used ports (in each line, after get_ports) according to the top level signal names in
5
6  ## Clock signal
7  set_property -dict {PACKAGE_PIN W5 IOSTANDARD LVCMOS33} [get_ports clk_a]
8  create_clock -period 10.000 -name sys_clk_a_pin -waveform {0.000 5.000} -add [get_ports clk_a]
9
10
11
12  ## Switches
13  #set_property -dict { PACKAGE_PIN V17 IOSTANDARD LVCMOS33 } [get_ports {in0[0]}]

```

```

162 create_debug_core u_ila_0 ila
163 set_property ALL_PROBE_SAME_MU true [get_debug_cores u_ila_0]
164 set_property ALL_PROBE_SAME_MU_CNT 1 [get_debug_cores u_ila_0]
165 set_property C_ADV_TRIGGER false [get_debug_cores u_ila_0]
166 set_property C_DATA_DEPTH 1024 [get_debug_cores u_ila_0]
167 set_property C_EN_STRG_QUAL false [get_debug_cores u_ila_0]
168 set_property C_INPUT_PIPE_STAGES 0 [get_debug_cores u_ila_0]
169 set_property C_TRIGIN_EN false [get_debug_cores u_ila_0]
170 set_property C_TRIGOUT_EN false [get_debug_cores u_ila_0]
171 set_property port_width 1 [get_debug_ports u_ila_0/clk]
172 connect_debug_port u_ila_0/clk [get_nets [list clk_b_IBUF_BUF6]]
173 set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe0]
174 set_property port_width 16 [get_debug_ports u_ila_0/probe0]
175 connect_debug_port u_ila_0/probe0 [get_nets [list {dout_b_OBUF[0]} {dout_b_OBUF[1]} {dout_b_OBUF[2]} {d
176 create_debug_port u_ila_0 probe
177 set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe1]
178 set_property port_width 16 [get_debug_ports u_ila_0/probe1]
179 connect_debug_port u_ila_0/probe1 [get_nets [list {din_a_IBUF[0]} {din_a_IBUF[1]} {din_a_IBUF[2]} {din
180 create_debug_port u_ila_0 probe
181 set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe2]
182 set_property port_width 1 [get_debug_ports u_ila_0/probe2]
183 connect_debug_port u_ila_0/probe2 [get_nets [list clk_b_IBUF]]
184 create_debug_port u_ila_0 probe
185 set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe3]
186 set_property port_width 1 [get_debug_ports u_ila_0/probe3]
187 connect_debug_port u_ila_0/probe3 [get_nets [list empty_OBUF]]
188 create_debug_port u_ila_0 probe
189 set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe4]
190 set_property port_width 1 [get_debug_ports u_ila_0/probe4]
191 connect_debug_port u_ila_0/probe4 [get_nets [list full_OBUF]]
192 create_debug_port u_ila_0 probe
193 set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe5]
194 set_property port_width 1 [get_debug_ports u_ila_0/probe5]
195 connect_debug_port u_ila_0/probe5 [get_nets [list ren_b_IBUF]]
196 create_debug_port u_ila_0 probe
197 set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe6]
198 set_property port_width 1 [get_debug_ports u_ila_0/probe6]
199 connect_debug_port u_ila_0/probe6 [get_nets [list rst_IBUF]]
200 create_debug_port u_ila_0 probe
201 set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe7]

```

- Implementation:

SourcesNetlist x ? _ □ □

FIFONets (245)Leaf Cells (101)dbg_hub (dbg_hub)u_ila_0 (u_ila_0)u_ila_1 (u_ila_1)

Properties ? _ □ □ x

Select an object to see properties

Project Summary x Device x ? _ □ □

Tcl ConsoleMessages x LogReportsDesign RunsPowerDRCMethodologyTiming ? _ □ □

☒ Error (3) ☒ Critical warning (3) ☐ Warning (11) ☐ Info (284) ☐ Status (545) Show All

Implementation (1 critical warning)

Route Design (1 critical warning)

[Timing 38-282] The design failed to meet the timing requirements. Please see the timing summary report for details on the timing violations.

Implemented Design (1 critical warning)

General Messages (1 critical warning)

[Timing 38-282] The design failed to meet the timing requirements. Please see the timing summary report for details on the timing violations.

Setup	Hold	Pulse Width
Worst Negative Slack (WNS): 3.113 ns	Worst Hold Slack (WHS): -0.141 ns	Worst Pulse Width Slack (WPWS): 3.750 ns
Total Negative Slack (TNS): 0.000 ns	Total Hold Slack (THS): -0.141 ns	Total Pulse Width Negative Slack (TPWS): 0.000 ns
Number of Failing Endpoints: 0	Number of Failing Endpoints: 1	Number of Failing Endpoints: 0
Total Number of Endpoints: 5613	Total Number of Endpoints: 5597	Total Number of Endpoints: 3221
Timing constraints are not met.		

Name	Slice LUTs (20800)	Slice Registers (41600)	F7 Muxes (16300)	Slice (815 0)	LUT as Logic (20800)	LUT as Memory (9600)	LUT Flip Flop Pairs (20800)	Block RAM Tile (50)	Bonded IOB (106)	BUFG (3)
FIFO	2001	3229	32	1015	1793	208	1147	2.5	39	
dbg_hub (dbg_hub)	483	739	0	228	459	24	323	0	0	
u_ila_0 (u_ila_0)	907	1495	27	475	784	123	517	1.5	0	
u_ila_1 (u_ila_1)	505	875	5	212	524	81	284	0.5	0	